Statistical Methods - Exercises

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1 Monty Hall Problem

Using the corollary to the Bayes Theorem, we get

$$
\begin{aligned}
P(F\ after\ switching) &= P(F\,|\,G)\,P(G) + P(F\,|\,F)\,P(F) = \\
&= 1 \times \frac{2}{3} + 0 \times \frac{1}{3} = \frac{2}{3} \\
P(F\ without\ switching) &= P(F\,|\,G)\,P(G) + P(F\,|\,F)\,P(F) = \\
&= 0 \times \frac{2}{3} + 1 \times \frac{1}{3} = \frac{1}{3} \qquad (1)
\end{aligned}
$$

where $P(X\,|\,Y)$ stands for the probability of winning X under the assumption that we have chosen the door with Y behind (with or without switching), while F and G stand for Ferrari and Goat, respectively.

2 The zero noise realization

Fixing the noise equal to zero while keeping its random nature (*id est* $\sigma_{noise} \neq 0$) we find that $Y_{obs} = Y_{true}$. In this way the noise does not introduce any shift over the parameter posterior distributions. However, since the zero noise realization is just one of the infinite possible realizations of a gaussian model of noise, the final parameters are still affected by uncertainties:

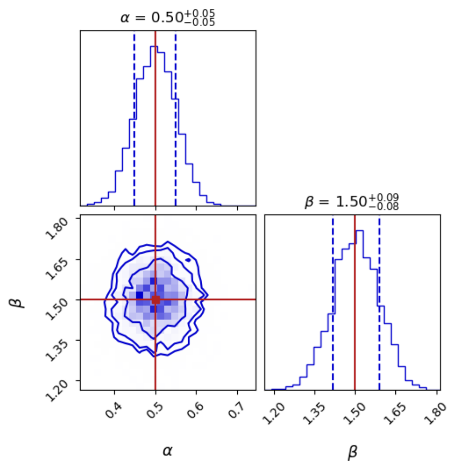

Figure 1: α and β corner plots with zero noise realizations and the same parameters used in the original notebook.

We can now show that averaging the log-likelihood over the non-zero noise realization corresponds to the log-likelihood computed in the zero-noise realization. Let us start by computing the average of n_i:

$$\begin{aligned} < n_i >= \int Dn'\, n'_i = \int \prod_{j=1}^{M} dn'_j \, \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{n'^2_j}{\sigma^2}} \, n'_i = \\ = \int \prod_{j\neq i}^{M} dn'_j \, \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{n'^2_j}{\sigma^2}} \int dn'_i \, \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{n'^2_i}{\sigma^2}} \, n'_i = \\ = 1^{M-1} \times 0 = 0 \, . \end{aligned} \qquad (2)$$

In the same way we obtain $< n_i^2 >= \sigma^2 \; \forall \, i = 1 \ldots M$.

Now we can average the log-likelihood over the non-zero noise realizations:

$$
\begin{aligned}
\log \mathcal{L}(d,\boldsymbol{\theta}) &= -\frac{1}{2\sigma^2}\sum_{i=1}^{M}(s_i - f(x_i,\boldsymbol{\theta})) + ... \\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{M}\left[n_i^2 + (f_{true} - f)^2 + 2n_i\,(f_{true} - f)\right] + ... \\
< \log \mathcal{L}(d,\boldsymbol{\theta}) > &= -\frac{1}{2\sigma^2}\sum_{i=1}^{M}\left[< n_i^2 > + < (f_{true} - f)^2 > + 2 < n_i >< (f_{true} - f) >\right] + ... \\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{M}\left[\sigma^2 + (f_{true} - f)^2 + 2 \times 0 \times (f_{true} - f)\right] + ... \\
&= -\frac{1}{2\sigma^2}\sum_{i=1}^{M}(f_{true} - f)^2 + ... \qquad (3)
\end{aligned}
$$

that is, the log-likelihood computed at $n_i = 0 \; \forall \, i = 1 \dots M$.